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Review article

A review of LoRaWAN performance optimization through cross-layer-based approach for Internet of Things

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ABSTRACT

The Internet of Things (IoT) is proliferating in technology and automation. Some popular IoT applications include environmental monitoring, home automation, agriculture, aquaculture, healthcare, transportation, and logistics. The construction of the IoT system is determined by where it will be used. The design can be tailor-made for short-range or long-range communication, rural or urban areas, indoor or outdoor applications, real-time or delay-tolerant communication, and much more. Low-Power Wide Area Networks (LPWAN) are frequently used for long-distance, energy-efficient, cost-effective communication. In LPWAN technology, Long-Range Wide Area Network (LoRaWAN) is one of the most popular choices because of its remarkable features and openness, making it highly suitable for IoT applications. However, despite its exceptional features, there are still ways to optimize the system to become more efficient and address several challenges during runtime. One possible way to address issues and challenges in LoRaWAN is through cross-layer optimization. This optimization technique violates the restrictions set by the Open System Interconnection (OSI) model and gives freedom to its protocol layers to communicate depending on the intended purpose. This paper surveys the state-of-the-art cross-layer approaches that optimize LoRaWAN for IoT applications. The cross-layer approaches were categorized according to the layer combinations and architecture. In addition, this paper provided observations on the effects of cross-layer optimization in LoRaWAN. Lastly, possible issues and solutions, challenges, and future directives from cross-layer optimization approaches were included.

1. Introduction

The utilization of the IoT is rapidly growing due to its usefulness in many aspects. These "things" have a unique identity in the system and can send or receive data with the help of the Internet through various networking protocols and applications [1]. IoT transforms conventional objects into being "smart" by manipulating their embedded technological capabilities [2]. The most common components of IoT are sensors, actuators, transducers, transceivers, and processing units [3]. Environmental monitoring, home automation, agriculture, aquaculture, healthcare, transportation, and logistics are some popular IoT applications [4]. IoT technology can automate information gathering, predict activities, monitor events, and analyze data without direct human intervention.

An IoT solution is configured depending on where it will be used. If it requires a high data rate with high throughput and reliability, a Wireless Local Area Network (WLAN) like Wi-Fi (802.11) is one of the popular choices. However, Wi-Fi is limited to

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short-range communication and has high energy consumption. On the other hand, LPWAN is the most popular option for applications requiring long-range, energy-efficient, and cost-effective communication. Among several LPWAN solutions, the remarkable features and openness of LoRaWAN technology stand out among others, making it highly suitable for IoT implementations [5]. However, no system can satisfy all the requirements of IoT applications. The top priorities for LoRaWAN are providing an adequate Quality of Service (QoS) and prolonging the network lifetime as long as possible.

Many factors in the LoRaWAN protocol stack can affect the network's lifetime and QoS. Understanding the process of each layer and how it affects the system can decrease energy consumption and optimize the system. The existing OSI model is highly efficient for wired networks. However, the results may become unfavorable when applied to wireless networks due to the strict processes between their layers. Cross-layer optimization is one of the techniques proven to improve the performance of wireless communications. The cross-layer optimization approach gives the layers in its protocol stack the freedom to communicate or collaborate with the other layers without restrictions [6]. This optimization process directly violates the conditions set by the traditional protocol, allowing the layers to interact in any way they want.

The major contributions of this paper are as follows:

1. Review of the state-of-the-art cross-layer-based approaches and designs implemented in LoRaWAN for IoT
2. Classification and discussions of optimization approaches in LoRaWAN for IoT based on cross-layer architectures
3. Identification of the issues, challenges, and potential solutions in the implementation of cross-layer-based optimization
4. Future research directions of cross-layer-based optimization for LoRaWAN in IoT

This review paper is an extension of the work in [7], which focused on classifying cross-layer-based approaches to optimize the LoRaWAN performance. The use of cross-layer in enhancing and optimizing wireless networks has proven effective. This technique can enhance network performance and QoS parameters independently or collaboratively with other computing processes. In addition, the cross-layer approach can offer an alternative way of determining how the layers can interact, depending on the algorithm defined.

The rest of the paper is organized as follows. Section 2 discusses the overview of LoRaWAN and cross-layer optimization for wireless networks. Section 3 reviews the related surveys focusing on LoRaWAN optimization and cross-layer optimization. Section 4 includes the search workflow, paper distribution, and bibliometric analysis. Section 5 discusses the classification of cross-layer-based approaches in LoRaWAN for IoT. In Section 6, the performance, advantages, and disadvantages of each cross-layer-based approach in optimizing LoRaWAN parameters are discussed. Section 7 presents the issues and possible solutions in using cross-layer optimization for LoRaWAN. Section 8 discusses the possible research directives for optimizing LoRaWAN by using a cross-layer approach. Finally, Section 9 concludes the paper.

2. Background

This section discusses LoRaWAN technology and cross-layer optimization for wireless networks.

LoRaWAN is in the class of LPWAN, where it is efficient in terms of energy utilization and addresses distance coverage challenges in IoT [8]. It can satisfy long-range communications while utilizing a small amount of energy. This technology has a straightforward application, making it easy to collaborate with other existing infrastructure, and can serve networks with low-power battery-operated IoT applications [9]. Most End Devices (EDs) depend on built-in and limited power supplies.

The operation of LoRaWAN is the same as that of other IoT devices, as shown in Fig. 1. The end devices gather data from the environment. The gateway will then receive the data gathered. The Network Server (NS) and the EDs are linked by the gateway (GW). The NS receives the data passed by the GW. It inspects if there are duplicated packets and relays the data to the Application Server (AS) [10]. The AS is responsible for processing application-specific data messages. The packets the AS receives will be forwarded to an application where the data is intended to be used [11].

The Open System Interconnection (OSI) model framework is based on a layered structure technique. The communication functions are partitioned into a vertical set of layers where each layer performs a set of functions [12]. The operation of the OSI model is based on the layered architecture and waterfall-like concept, as shown in Fig. 2. The traditional LoRaWAN network adopts the OSI model framework for communication.

The network standards may only define several of the layers in the OSI model due to the characteristics of wireless networks [13]. For example, the protocol stack of the LoRaWAN technology is shown in Fig. 3.

Adopting the OSI model structure for wireless communication makes it difficult for the network to adapt to environmental changes [14]. The sequential processing of this model cannot provide fast accessibility of information in every layer in the stack [15]. In addition, transfer overheads incurred by every layer limit the interoperability of the system [16]. One known process in wireless communication optimization is the cross-layer approach. This optimization technique can improve network lifetime and enhance the QoS of the system. It can also be an alternative to overcome several issues, like overheads, and could efficiently increase the usability of information shared among the layers [16]. Therefore, a cross-layer approach is highly beneficial for systems with limited resources.

The main idea of the cross-layer is to create a link between the different processes in each layer and minimize the overhead incurred in the stack [17]. One of the definitions of cross-layer is the violation of a layered communication architecture to the original architecture [6]. This optimization technique can correlate the different layers in the network. For example, there are three layers: Layer 1 (L1), Layer 2 (L2), and Layer 3 (L3). With cross-layer, L1 and L3 can communicate and know each other's information.

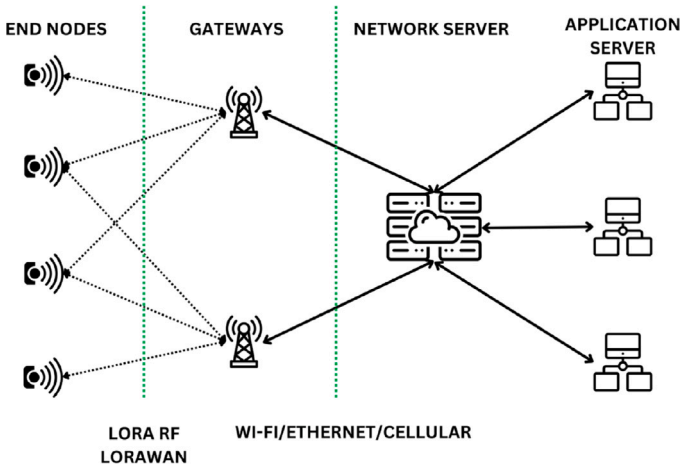

Fig. 1. LoRaWAN architecture.

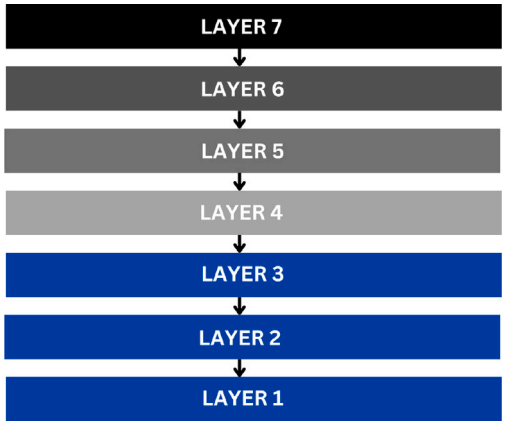

Fig. 2. Layered architecture model.

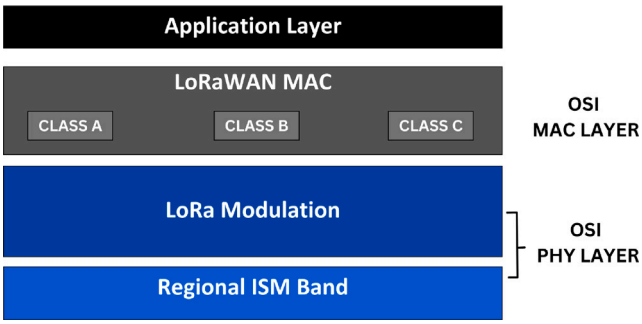

Fig. 3. LoRaWAN protocol stack.

L1 and L2 can merge into a "super layer" and share their parameters and data. Another option is that L1, L2, and L3 can share and access their data back and forth.

These layers can interact and optimize the LoRaWAN system regarding energy efficiency, communication range, or multiple access through different cross-layer approaches. There are many available cross-layer categories in the literature, one of which categorizes the cross-layer based on its architecture [6]:

1. *Merging of Adjacent Layers*: The merging of layers is designed to collaborate on the different services the respective layers provide. A visual representation of this cross-layer architecture is illustrated in Fig. 4. This architecture can combine two or

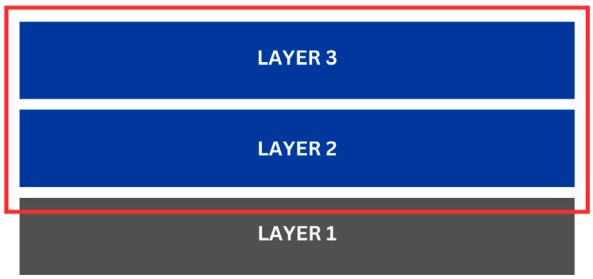

Fig. 4. Merging of adjacent layers.

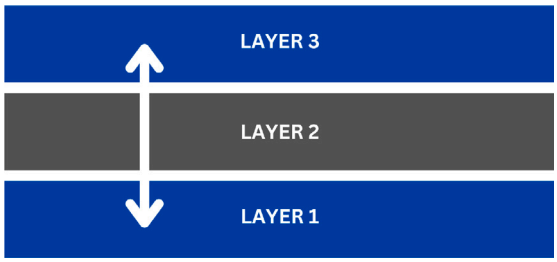

Fig. 5. Direct communication of layers.

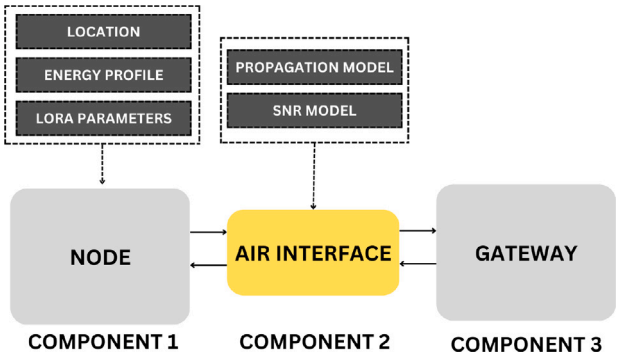

Fig. 6. Completely new abstraction.
Source: Adopted from [18].

more layers. This usually creates a protocol to control the routing and scheduling of the network collaboratively. However, this approach can introduce network latency depending on the routing and scheduling protocol selection.

2. *Direct Communication of Layers*: This architecture allows information sharing between adjacent or non-adjacent layers, as illustrated in Fig. 5. This architecture only utilizes several parameters from the different network layers. Compared to the merging of layers, this architecture does not entirely alter the protocol stack since most of the processes are still adapted. However, the layers' interoperability should be considered since other layers have no relationship.
3. *Completely New Abstraction*: This architecture proposal can manage the layers in heaps and not in stacks or smaller units called building blocks, as shown in Fig. 6. This approach can introduce flexible, precise, and fast interactions, depending on the processes employed. However, several conditions and considerations must be set since the protocol stack is completely altered.
4. *Shared Database*: A common database that all the layers can access is another architecture proposal. This database can be seen as another layer that can store and process information in all the layers, as illustrated in Fig. 7. It could be a controller or a bridge that the layers in the stack can access. However, as an additional layer, a database can introduce complexity to the design since it is incorporated with other mechanisms or powerful computational software.

The MAC and PHY layers are the major components of the LoRaWAN protocol stack. LoRa is the PHY layer that utilizes a Chirp Spread Spectrum (CSS) modulation technique and carries the data [19]. On the other hand, LoRaWAN defines the network architecture, communication protocols, operating frequencies, and nature of the connected devices. The application layer is not

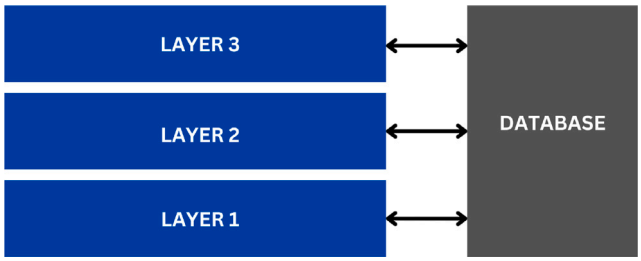

Fig. 7. Shared database.

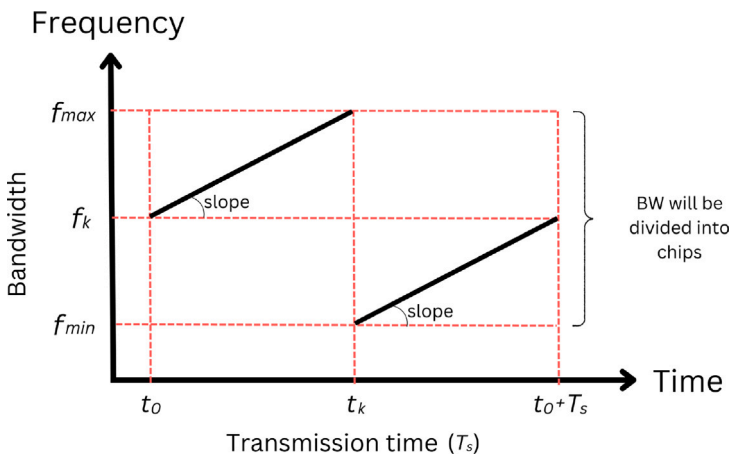

Fig. 8. CSS modulation used by LoRaWAN.

defined in the LoRaWAN specification. However, the author of [20] identified several platforms used for LoRaWAN, like Loriot, IBM Bluemix, Amazon, and AWS. These layers perform independently and are discussed as follows:

2.1. Physical layer

LoRaWAN uses the PHY layer called LoRa based on CSS. This technique is based on varying frequency for time [21] as shown in Fig. 8. The CSS mechanism converts raw information called symbols into a sweep signal or chirps [22]. The CSS symbol depends on four parameters: Spreading Factor (SF), the frequency range (f_{min} and f_{max}), and the input bits [23].

For example, SF8 has eight symbols, meaning the symbol can hold 2^{SF} chips, in this case, 256 chips. Imagine a symbol equal to 10000000 (SF8); this will have a decimal value of 128. A sweep signal from f_{low} to f_{max} will be divided into 256 chips. The starting frequency (f_k) will start at 128. The signal will constantly chirp starting from f_k until it reaches f_{max}, then will continue from f_{min} to f_0. The authors of [21] derived the duration of one symbol, where it can be computed as:

$$T_s = \frac{2^{SF}}{BW} \tag{1}$$

The slope of the frequency variation can be computed as follows:

$$slope = \frac{BW}{T_s} \tag{2}$$

The symbol k represents a binary number with equal length to the SF. This defines the initial frequency f_k and shrink point t_k [21].

The LoRa PHY layer has several tunable parameters to define performance and energy utilization [24]. These parameters are often optimized to manage the system's limited resources efficiently, and they are as follows:

1. *Transmission Power (TP):* The LoRa TP with steps of 1 dB can be tuned from −4 dBm to 20 dBm. Due to the implementation limits of the hardware, the standard range is from 2 dBm to 20 dBm [24]. The limit of the power controls is only 2, 5, 8, 11, 14, and 20 dBm. The TP also depends on the regulations imposed in a particular region. For example, in the EU 863–870 band, 14 dBm is the maximum transmission power that can be utilized [11]. Utilizing high transmission power does not automatically mean it produces a better communication system. A high TP can minimize the sound-to-noise ratio (SNR) but requires the ED to consume additional energy [25].

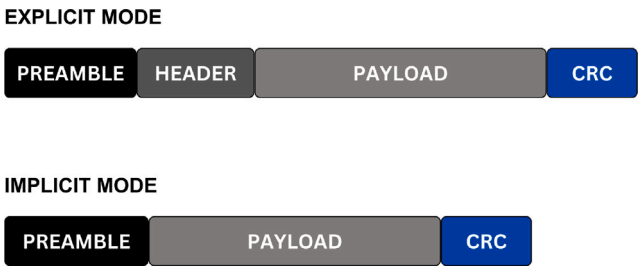

Fig. 9. LoRa packet modes.

2. *Bandwidth (BW) and Carrier Frequency (CF):* The BW of a LoRa network primarily utilizes 125 kHz, 250 kHz, or 500 kHz [11]. Employing a higher BW can increase the rate of data transmission. However, the increase of BW can also increase the probability of picking up additional noise in the channel, thus affecting the receiver sensitivity. The CF can be configured from 137 to 1020 MHz. The frequency range that can be used depends on the regulations imposed on certain regions [11].
3. *Spreading Factor (SF):* The SF is one of the most essential configurable parameters in a LoRa device [24]. The SF values of LoRa can utilize values of 7 to 12 (SF7–SF12) [11]. A symbol is the raw bits of data identified by the SF. For example, an SF of 8 (SF8) has a symbol of 8 raw bits. And a symbol of 8 bits can hold 2^{SF} (2^8) 256 chips.
The SF can also be viewed as
$$SF = log_2(\frac{R_c}{R_s}) \tag{3}$$
where R_c and R_s are the chip rate and symbol rate respectively [11]. The chip rate R_c is also the same as the BW. For example, a bandwidth of 250 kHz has a chip rate of 250,000 chips/s. The symbol rate (R_s) can be computed as
$$R_s = \frac{BW}{2^{SF}} \tag{4}$$
The SF also greatly impacts the data rate (DR). The DR can be computed as:
$$DR = (SF)(R_s)(\frac{4}{4+CR}) \text{ bits/s} \tag{5}$$
An increase in SF will decrease the DR. A higher SF can increase the signal-to-noise ratio (SNR), range, and sensitivity [24]. However, it will also increase the time it takes to transmit packets, thus consuming additional energy. LoRa transmissions are orthogonal, meaning the gateway can receive several packets simultaneously if it has different SF [24]. Due to the different SF adopted by LoRa, this parameter is commonly included in optimization processes.
4. *Coding Rate (CR):* The CR is the proportion of actual data bits that carry the information. A mechanism for error control called Forward Error Correction (FEC) creates error correction bits included in the transmitted data. The CR can be set to 4/5, 4/6, 4/7, or 4/8 [24]. With CR set to 4/8, half of the bits are the actual data, and the other half are the redundant bits. Utilizing additional error correction bits offers more protection to the data but requires a longer transmission time [24].

A LoRa packet comprises a preamble, header, payload, and CRC field [26]. The LoRa packet can either be in explicit or implicit mode, as shown in Fig. 9. The explicit mode contains a header with information on the payload length, coding rate, and if a cyclic redundancy check (CRC) is present [11]. The implicit mode does not contain a header since it has a fixed payload and coding rate. Excluding the header can reduce the transmission time [26]. A preamble is used for synchronization and is composed of 8 successive up-chirps. The payload field comprises the actual data, a MAC header, and MIC. The actual payload can be from 2 to 255 bytes. The MAC header identifies the frame type as an acknowledgment or data. The MIC is the digital signature of the payload [26]. Lastly, the CRC field is optional and used for error protection of the payload [26].

In addition, the Time-on-Air (ToA) is the time it takes to transmit a packet until it reaches its destination. ToA can be defined as [26]
$$ToA = T_{preamble} + T_{payload} \tag{6}$$
where $T_{preamble}$ is the preamble duration. LoRa recommends a preamble of 8 bits to be used in all regions. Due to hardware implementation, it includes an additional 4.25 bits. Therefore, $T_{preamble}$ can be computed as
$$T_{preamble} = (n_{preamble} + 4.25)T_s \tag{7}$$
where $n_{preamble}$ is the preamble length and T_s is the time of one symbol. For the time of the payload $T_{payload}$, it can be computed as
$$T_{payload} = (PLSymbol)(T_s) \tag{8}$$
$$T_{payload} = 8 + max(ceil(\frac{8PL - 4SF + 28 + 16CRC - 20H}{4(SF - 2DE)})(CR + 4), 0) \tag{9}$$
where:

- PL — number of payload (bytes)
- SF — is the Spreading Factor (7–12)
- CRC — when enabled = (1), disabled = (0); the LoRa default is (1)
- H — when enabled = (0) meaning in explicit mode, disabled = (1) meaning in implicit mode
- DE — when enabled = (1), disabled = (0)

This shows that the transmission time of packets depends on the selected parameters. The longer it takes to send a packet, the more energy the end devices consume. Therefore, correctly selecting the SF and CR greatly influences LoRa transmission ToA [26].

LoRa GW and EDs strictly abide by the DC requirements imposed by certain regions when utilizing the unlicensed ISM Band. The DC is implemented for EDs to maintain fairness in accessing the channel. LoRa, based on EU standards, recommends a DC of 1% [27]. The device must wait 99 times the time it used during its last transmission. This is referred to as time-off (T_{off}) and can be calculated as:

$$T_{off} = \frac{ToA}{DC} - ToA \tag{10}$$

where ToA is the Time-on-Air and DC is the duty cycle imposed by a specific region.

LoRa modems also provide a mechanism called Channel Activity Detection (CAD). This is implemented to detect LoRa preambles in the channel quickly. There are two phases for CAD: the reception phase and the processing phase [27]. In the reception phase, LoRa ED switches to receive mode using a pre-configured SF and captures all the symbols in the channel. In the signal processing phase, the LoRa modem finds a relationship between the received waveform symbols and the ideal waveform of the preamble symbols. If the results match, the device can switch to receive mode and receive the payload [27]. The use of CAD can also contribute to the energy consumption of the EDs depending on the SF and BW. Therefore, a proper allocation and combination of the parameters should also be considered when enabling CAD [28].

The LoRaWAN network also employs Adaptive Data Rate (ADR). The purpose of ADR is to efficiently configure the transmit power and data rate of the end devices [29]. This optimization maintains connectivity while using the smallest ToA possible [30]. The ADR mechanism is performed in the EDs and the NS. The network can modify the TP and decrease the SF in this algorithm. On the other hand, the EDs can only increase the SF [29]. The network server collects the 20 recent transmission data points of the end nodes. Then, the network server will assess and optimize the parameters based on the collected information. The ADR changes the data rate depending on the link budget. If the link budget is high, the data rate will increase. On the other hand, if it has a lower link budget, the data rate can be decreased [31]. The ADR mechanism is preferred to be used for static EDs. Every transmission will provide different parameters if the ED is dynamic or mobile (e.g., SF and SNR). The collected transmission data at the NS will vary greatly in SF and SNR. Therefore, this limits the ADR's ability to select appropriate parameters for the ED.

2.2. Link layer

The LoRaWAN MAC is the equivalent of the OSI Model Link layer. LoRaWAN is a software-based layer that operates above the LoRa PHY layer. This layer controls the state (i.e., active and sleep) of the EDs in the network. The proposed MAC protocol should manage a proper trade-off between network lifetime, reliability, fairness, scalability, and latency [32]. An ALOHA-type protocol is used for transmission slot schedules of the end devices [33]. LoRaWAN has three classes, Class A, B, and C, and all initiate an ALOHA manner during uplink transmission [30]. However, these classes have a different downlink receiving window schedule. LoRaWAN Class A will open two receive windows (RX1 and RX2) to wait for a response from the NS. The first receive window (RX1) will open for a certain time (T1) to listen for incoming messages from the NS. The second receive window (RX2) will open for a specific time (T2) after the uplink transmission using a default channel [34]. For Class B, end devices provide additional receive windows for the downlink transmission. After the transmission, a beacon window will open and be divided into ping slots [34]. Class C end devices open a short-second receive window (RX2) with a time (T1) before RX1. After which, RX1 will open for a specific amount of time (T2). After T2 has ended, RX2 will continuously open up until the subsequent uplink transmission of the end device [34].

LoRa EDs can adapt to different MAC protocols. The two main approaches are the reservation-based and contention-based protocols [35]. Time Division Multiple Access (TDMA) is an example of a reservation-based protocol. The time is divided into frames, and frames are further divided into slots. The EDs can only transmit data during the assigned slot. This guarantees a predictable scheduling delay and increased throughput in the network since collisions are prevented [36]. On the other hand, ALOHA is an example of a contention-based protocol. The EDs compete for the use of the medium; for example, the node closest to the receiver or with the highest power has a higher chance of successful transmission. Contention-based protocols are much easier to implement because information about the topology and synchronization is not required compared to reservation-based protocols [35].

LoRaWAN uses an FEC mechanism to help restore the data in case it gets corrupted, where it systematically generates redundant bits. This allows the receiver to correct the identified errors in the received data [31]. An ED can use a 4-bit CRC header and a 16-bit CRC at the end of the payload for error detection. The CRC is used to detect error codes in corrupted data.

2.3. Network layer

The NS of a LoRaWAN network processes the routing, management, and monitoring of the devices [37]. LoRa can also adapt different routing protocols depending on the application. Ideally, routing protocols often select a suitable path for the data to travel. However, the routing process in wireless networks often encounters difficulties due to network conditions, channel characteristics,

Table 1
Summary of state-of-the-art survey and review papers in LoRaWAN optimization.

Paper	Topic	Focus	Contribution
[11]	Energy efficiency optimization in every layer of LoRaWAN	Single and multiple parameter configurations, new scheduling protocols, or replacing the existing routing mechanism	Possible research path to optimize LoRaWAN and a comprehensive LoRaWAN tutorial
[27]	LoRaWAN communication protocols under energy efficiency perspective	Routing and scheduling protocols based on an energy efficiency perspective	Critical aspects on LoRaWAN communication protocol focusing on energy efficiency and a proposal of a Green LoRaWAN communication protocol for network lifetime and network QoS for harsh environments
[39]	Research problems in LoRaWAN	Solutions identified are mostly based on single-layer optimization and hardware-related solutions	Taxonomy of LoRa research challenges
[40]	LoRaWAN challenge, research issues, and state-of-the-art LoRaWAN based protocols	General discussion on different MAC protocols	Identification of LoRaWAN based MAC protocols research issues
[41]	Long-range wide-area network technology optimizations	Systematic review of state-of-the-art works for LoRaWAN optimization solutions for IoT networking operations	A taxonomy classification to categorize the aspects that affect the LoRaWAN operation, the co-existence of IoT devices and cited applications, resource allocation techniques, MAC layer, network planning, and mobility solutions
[42]	LoRaWAN recent trends, opportunities, and simulation tools	Different ADR modifications, mathematical optimization models, scheduling schemes, and hardware selections	Identification of different mathematical optimization models and LoRaWAN simulation tools
[43]	Survey on ADR optimization in LoRaWAN	Different ADR proposals to address scalability, throughput, and energy consumption	Comparison and discussion on different ADR modifications based on scalability, throughput, and energy efficiency
[44]	Viability of confirmed traffic in a LoRaWAN	Different studies showing the viability of confirmed traffic in LoRaWAN and how it can affect network performance.	Challenges and possible solutions for confirmed traffic
[45]	Survey on scalable LoRaWAN for massive IoT	Possible solutions to signal interference and collision for concurrent transmissions, use case/applications for LoRaWAN	Identified solutions for signal interference and collision, use case applications for massive IoT, and future research directives
This work	A survey on cross-layer optimization in LoRaWAN for IoT	Different cross-layer optimization approaches done in LoRaWAN for IoT	Classification and performance evaluation of cross-layer optimization in LoRaWAN for IoT, identified advantages and disadvantages of using cross-layer optimization in LoRaWAN, issues, challenges, and future directives

and performance metrics [38]. One of the categories of routing protocols is route discovery-based routing. The routing protocols for route discovery are either reactive or proactive. Reactive protocols create routes on demand when queries are initiated. Proactive protocols are table-driven, and all routes are computed before implementation.

The following section discusses the survey papers that addressed several issues and challenges for LoRaWAN systems and cross-layer optimization for wireless networks.

3. State-of-the-art surveys and reviews in LoRAWAN performance optimization

The optimization of the network lifetime, communication range, and multiple access are still outstanding issues despite the remarkable performance of LoRaWAN. Many factors can affect the energy utilization of LoRaWAN end devices. It can be through the collision of data, inefficient tuning of configurable parameters, routing, or scheduling processes, to name a few. The end devices must conserve and efficiently manage their limited resources as much as possible. On the other hand, addressing issues related to communication range should consider several factors like efficient routing and determining environmental factors. The catering of thousands of end devices to address scalability is also a challenge for IoT applications. In line with this, the following discusses the survey papers that focus on optimizing LoRaWAN and cross-layer optimization in wireless networks. The summary of these papers can be seen in Table 1.

The optimization of energy utilization in LoRAWAN is the focus of the papers of [11,27]. The work of [11] is based on energy efficiency optimization in all the layers (PHY, LINK, and NET) and mainly focuses on the network characteristics. Most solutions are

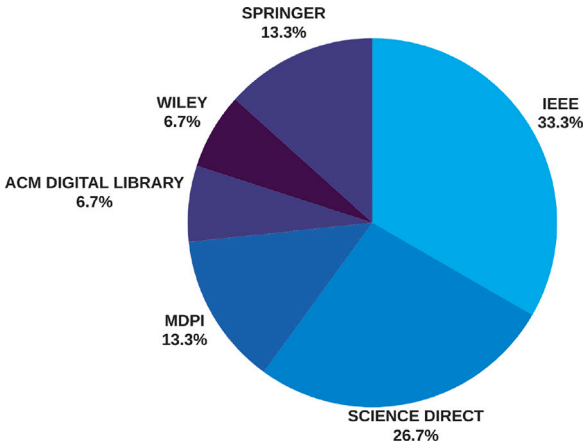

Fig. 10. Paper distribution based on database sources.

based on single and multiple parameter configurations, new scheduling protocols, or replacing the existing routing mechanism. For the work of [27], the authors discussed the critical aspects and research challenges of the LoRaWAN communication protocol. The categorization of the possible solutions cited by the authors is directed toward multi-access, routing, and energy-efficient protocols.

The focus of the authors of [39–41], and [42] is directed towards identifying possible solutions that can address LoRaWAN challenges in general. The authors of [39] categorized the cited solutions based on the devised taxonomy of LoRa research challenges. The taxonomy includes challenges regarding energy consumption, communication range, multiple access, error correction, and security. The solutions to these challenges are primarily based on single-layer optimization and hardware-related. For the authors of [40], the direction of their paper is to cite several MAC protocols related to resolving issues regarding scalability, throughput, coverage, and energy conservation. In the work of [41], the authors created a taxonomy of LoRaWAN technology optimization based on the co-existence of IoT device applications, resource allocation, MAC layer, network planning, and mobility. In addition, the authors also included network solutions based on network slicing and multiple radio access technology (Multi-RAT) in their work. The authors of [42] evaluated recent approach solutions and presented methods to address power consumption, pure data extraction, scalability, coverage range and QoS, and network security.

There are also survey papers for LoRaWAN optimization based on specific topics, like the papers of [43,44]. The authors of [43] focused their study on the different LoRaWAN ADR modification proposals. The ADR enhancements addressed scalability issues, throughput, and energy consumption for LoRaWAN networks. Their significant contribution was comprehensively analyzing solutions developed to optimize ADR algorithms. Lastly, the authors of [44] focused on exploring the viability of confirmed traffic in LoRaWAN. Their study included cited protocols and applications where the PDR is improved through acknowledgments.

These survey papers indeed contributed to studying the different optimization schemes of LoRaWAN networks. However, none of these papers explicitly discusses cross-layer optimization for LoRaWAN networks. To our knowledge, no survey papers about cross-layer optimization in LoRaWAN for the IoT have been published. Therefore, this paper presents a survey on the performance of state-of-the-art cross-layer approaches used to optimize the performance of LoRaWAN in IoT. This study also discusses the issues, challenges, possible solutions, and future directives for cross-layer optimization in LoRaWAN networks. The summary of the review papers related to LoRaWAN optimization is shown in Table 1.

The following section discusses the selection process for the cross-layer optimization papers used in this work.

4. Search workflow and paper distribution

This section discussed the methodology used in conducting the search workflow and paper distribution of the existing cross-layer optimization for LoRaWAN in IoT. The methods used in gathering the papers used in this survey are the following:

- *Searching*: The keywords used in searching for cross-layer-based LoRa optimization papers are LoRa cross-layer, LoRaWAN cross-layer, LoRa optimization, and LoRaWAN optimization. The databases used are IEEE Xplore, ScienceDirect, MDPI, Springer, Wiley, and ACM Digital Library. Several papers in Springer and Wiley discuss either cross-layer approaches or LoRaWAN. However, these papers do not focus on parameter optimization for LoRaWAN.
- *Paper selection*: This part focused on selecting the papers that conducted cross-layer in the network and optimized one or several LoRaWAN parameters. The papers are selected based on the cross-layer architecture used (i.e., within the LoRaWAN stack) and the protocol's effectiveness. The percentage of databases and the publication year of the papers used in this survey are shown in Figs. 10 and 11, respectively. 11 research articles and approaches published from 2018 to 2023 were reviewed critically in this paper.

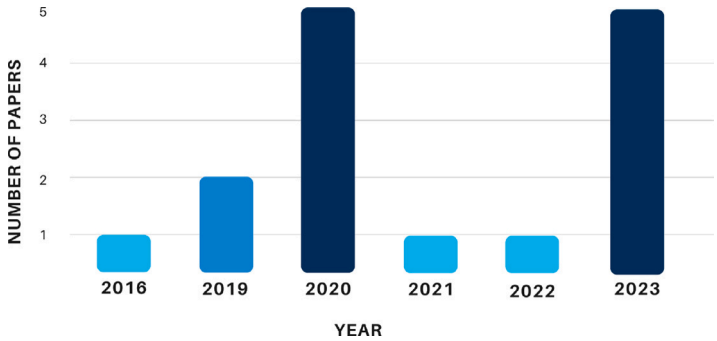

Fig. 11. Paper distribution based on publication year.

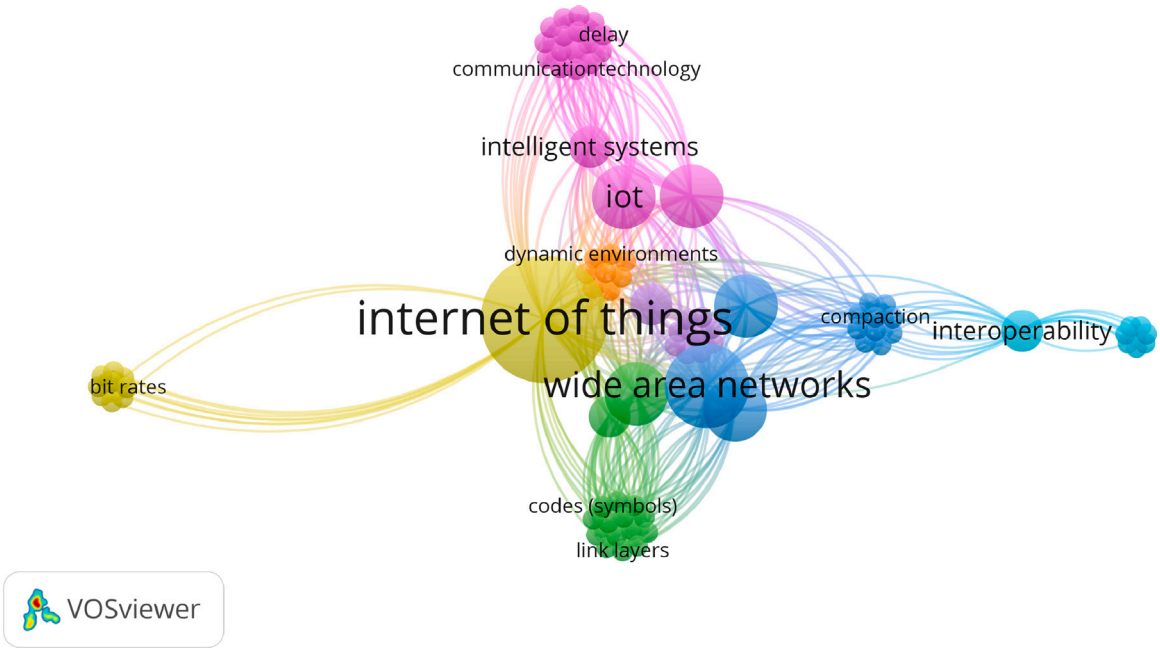

Fig. 12. Network keyword analysis for "LoRaWAN", "Internet of Things", and "cross-layer".

- *Discussions*: This part gathered critical information from the selected papers to identify how cross-layer optimization affects the LoRaWAN system. The basis of the discussions is the cross-layer architecture used, the purpose of crossing the layers, and the optimized parameter. This was used to identify the effects of cross-layer optimization in LoRaWAN regarding energy utilization, routing and scheduling, and the impact of the end node topology. Lastly, a discussion on the advantages and disadvantages, issues and challenges, and future directions for cross-layer optimization in LoRaWAN for IoT was included.

The Scopus-indexed keywords "LoRaWAN", "Internet of Things", and "Cross-layer" generated the bibliographical visualization using VOSviewer [46] as shown in Fig. 12. The system generated 86 keywords with a minimum number of occurrences. In the analysis, the searched keywords showed strong links and topics related to wide area networks, IoT, intelligent systems, interoperability, communication technology, delay, dynamic environments, codes, link layers, compaction, and bit rate.

5. Review and classification of cross-layer approaches in LoRaWAN for IoT

This section presents the different approaches of cross-layer architecture for LoRaWAN optimization. The classification of the papers is based according to the cross-layer architecture used in the optimization process. These classifications are merging adjacent layers, direct communication of layers, completely new abstraction, and a shared database [7].

5.1. Merging of adjacent layers

According to [47], the involvement of the network and link layers tends to address specific issues by modifying the routing and scheduling protocols. In this survey, all cross-layer proposals merged the Link and Network layers. This cross-layer approach tends to perform scheduling and routing simultaneously or dependent on each other.

The authors of [28] integrated the Link and Network layer to achieve multi-hop communication and time synchronization among nodes to define the slotted channel access. Every beacon sent by the GW is used to synchronize the EDs. This also indicated the start of an epoch, where an epoch contains N slots. Synchronization between EDs and the GW is achieved through the flooding of beacons. Connected EDs in the GW must listen and wait for the beacon to be received before data transmission can occur. On the other hand, channels are virtually divided into time slots every beacon period. The transmission can only commence during the start of every time slot. This scheduling can mitigate the collision of packets in contrast to the existing ALOHA protocol used by LoRaWAN. End nodes not within the gateway's range can still receive and transmit updates through their neighboring nodes. According to the test bed results, this protocol can deliver packets from all the nodes up to 80% reliably.

The protocol named JMAC [48] merged the data link and network layers for flexibility in supporting different IoT applications. The cross-layer approach was used to predict the wake-up time of end devices and support multi-hop routing at the same time. The core idea of this protocol is for the nodes to communicate with their neighboring nodes regarding their active time and distance through hops. The routing, scheduling, and identification of the state (awake or sleep mode) rely on how the nodes interact. The goal of this protocol is to keep the end nodes sleeping as long as possible until they are required to relay or transmit a message. The nodes operate in two modes: to discover neighboring nodes and to implement the JMAC protocol. The nodes periodically announce their information using beacons when they join the network. After this, they wait the rest of the time, expecting to receive announcements from their neighboring nodes to update the topology. After synchronizing the nodes, every node opens a receive window for the child node's data. That data will be put in a queue and sent as an ACK message to the child. When no data is received during a specified timeout, the nodes return to sleep and schedule a wake-up time. The MAC protocol used in this study was inspired by the Asynchronous Scheduled MAC (AS-MAC) and Receiver-Initiated MAC (RI-MAC) protocols. The AS-MAC is a protocol where nodes know their neighbors' wake-up time and do not require long preambles before starting their transmission [49]. Meanwhile, the Receiver-Initiated MAC (RI-MAC) is a protocol where the sender and receiver find a specific time to exchange data but are still independent with their duty-cycle schedules [50]. This study showed a better PDR when there were more parents to balance the traffic. In addition, the simulation results concluded that this protocol allows end devices to consume less energy and, at the same time, extends the coverage of baseline LoRa technology.

The study of [51] aims to minimize latency and increase reliability in multi-hop networks. This proposal is somehow similar to the concept of [48]. The key features of this work are distributed aggregation scheduling, parallel transmission to minimize latency and time slots, and minimized packet size. Data aggregation integrates the subset of information to reduce the data volume [52]. The authors of [51] exploited this feature to assign collision-free cells to neighboring nodes. Moreover, this allows the nodes to create a schedule based on their neighbors' information without requiring network topology information. This work used data aggregation for child nodes to send their data to their parent nodes before the parent sends it. Data aggregation is often implemented in a tree topology connected to the sink [53]. In this case, a tree construction algorithm maximizes the channel to achieve a high balance load. A time slot and channel assignment algorithm were used for the scheduling process to support the intended parallel transmission. A Time Division Multiple Access was used to guarantee no collision in the nodes during tree construction. The nodes need to compete with other nodes to get accepted in the network and have a slot for transmission. If the network determines there are still nodes trying to join the network, the gateway can send a command to add the nodes for it to join. This protocol showed a high probability of data reception. In addition, the parallel transmission reduced the number of time slots and the end-to-end delay transmission of packets.

A Multi-hop network based on the Distance Sequenced Distance Vector (DSDV) routing protocol is proposed by the authors of [54]. A cross-layer approach was used to identify the Routing Node's (RN) decision to transmit data depending on the network conditions. This protocol has three threads operating in a parallel manner. A Beacon Thread configures the channel parameters for the reception of the beacon. The second thread is a timer that can implement "interrupts" when it detects a time-out or error in the received packet. It can also indicate the start of the receiving window. Third is a transmission thread, which can manage the transmission of the full dumps and data packets. It also manages the subsequent transmission based on the calculated time on air and the duty cycle it needs to abide by. A Data Reception Thread's purpose is to manage all the received packets except the beacons sent by the gateway. The end nodes store a routing table indicating the next reachable hop destination, the number of hops to reach that destination, the assigned sequence number, and a time stamp that its destination node will assign. The DSDV routing protocol can implement a full dump or incremental packets as routing messages. A full dump carries all the entries indicated in the routing table, while an incremental packet only carries the changed information based on the last full dump received packet [55]. The nodes not in the gateway's range transmit their message to an RN until it reaches the gateway. The message of the Leaf Node (LN) is sent to the gateway based on the routing information exchanges of the RNs until it finds a route. The LN does not participate in the routing process since the authors assumed these were energy-constrained nodes. The LN only needs to broadcast its routing updates regularly. The LN's path destination is guaranteed due to its contained Unified ID field. After constructing the route, the LN can transmit a LoRaWAN packet with additional routing overhead to the RN. Fortunately, this overhead will be dropped once it reaches the RN. This allows the application server to receive only the LoRa packet and information on what node transmitted that message. On the other hand, the gateway sends a beacon at regular intervals so that the RNs can update their routing table and perform a full dump as soon as possible. RNs who did not receive a beacon can receive a full dump from their neighboring RNs.

Table 2
MAC and routing protocols used in the merging of layers.

Paper	MAC protocol	Route discovery
LoRaBlink [28]	Reservation-based	Proactive
JMAC [48]	Reservation-based	Proactive
Mai et al. [51]	Reservation-based	Proactive
Dias et al. [54]	Reservation-based	Proactive
OppLoRa [56]	Reservation-based	Proactive
IRIS [60]	Reservation-based	Reactive

An algorithm iterates the received full dump and checks the changes in the data. The routing tables can be updated based on new destinations, better network metrics, or newer sequence numbers. The RNs that are neighbors with the gateway schedule a short receiving window after they receive a beacon due to the changes in the channel parameters for the parameters used by the gateway in transmitting the beacons. The other RNs also schedule their full dump based on the next instance, where they expect the first full dump of the next beacon period.

The authors of [56] proposed a protocol called OppLoRa inspired by Opportunistic Routing for Wireless Mesh Networks (WMN). This protocol aims to improve the end-to-end reliability of LoRa significantly. OppLoRa is based on a modified version of ExOR [57] and Distance Sequence Distance Vector (DSDV) route discovery. The protocol ExOR integrated the routing and MAC to improve the routing through the broadcast nature of the multiple transmission opportunities [57]. Opportunistic Routing (OR) sends packets to a node, allowing neighboring nodes to overhear the broadcast [58]. The nodes would be considered forward relay sets if they overheard the broadcast [59]. The candidate nodes with the smallest hops to the destination are the basis for selection. Therefore, OR mainly selects the forwarder set and prioritizes the nodes in the set. OppLoRa also extended the ALOHA protocol of LoRaWAN by including slotted acknowledgments for data frame receptions and coordination. It supports the OR by providing a reliable anycast transmission to the selected next-hop CS on the data link layer. One of the simulations evaluated OppLoRa in a cross-layer manner, where the link layer can return the packet to the network layer if the transmission fails. The purpose is for the network layer to select different forwarders for re-transmission. With the increase in candidates in the forwarders set, the cross-layer approach is more advantageous than traditional LoRa in terms of end-to-end packet error ratio.

One specific regulation for using the unlicensed ISM band is to abide by the DC regulation. The IRIS [60] protocol intends to further lessen the DC of EDs to less than 1%. IRIS integrated the network and MAC to utilize simple node logic by sending ping packets to set up routes and slotted schedules. A pipeline type of network inspired the topology used in IRIS. The EDs are randomly distributed along a pipeline where two base stations (BS) are connected end to end. A start (BS) sends ping packets to the EDs within the network until it reaches the End BS to determine the route. The MAC layer configures the active and sleep states of the EDs. The active state is divided into transmission and reception modes. The time of one duty cycle is referred to as a frame, and the frame is further divided into slots. This ping packet transmission's timing defines the start of the frame timing. The route nodes are the EDs that can relay ping packets, while non-routed nodes are the ones that report monitoring information to the route nodes. The IRIS protocol only requires a unique ID code for EDs to find its route. The EDs only require a unique ID and a pre-programmed slot duration, number of slots per frame, and number of slots required to be active. The transmitted ping packet indicates the start of a frame. The ping packet contains an integer packet type (4 bits), integer IDs of the sender and destination (next-hop) nodes (11 bits), a binary link ID to indicate if a route is formed (1 bit), and a reserved payload. The sender can note "−1" in the link ID if the sender does not have an intended destination node. This indicates that the ED is looking for a next hop to join the network. The EDs wake up randomly and listen accordingly to the defined active period, then return to sleep mode. The ED delays its active period if it does not hear anything while awake. When an ED hears a ping with "−1", it replies with an ACK to declare its interest in being a route ED. The ED configures its active period in the next frame based on the last ping it received. The ED configures its active state to begin with a random slot in the next frame if the ID is still marked as "−1". It will be considered a non-route ED if the destination ID is not "−1" or its ID. If the destination of the ping packet is its ID, then the sender has selected the ED as the next hop, making it a route node. The new route node repeats this process until it reaches the end BS. The route can be successfully constructed if the end BS informs the start BS that it already received the ping.

The following studies used a reservation-based protocol similar to TDMA. The use of this protocol can introduce fairness in the transmission, reduce collision, and address multiple access challenges. However, the network topology and synchronization information require overheads [36]. On the other hand, a proactive route discovery was used to maintain the best path and avoid routing loops. However, the routing table should be often updated. Using cross-layer, merging these layers minimized the overheads incurred in the scheduling and routing process. For [60], a reactive-based route discovery approach was used. The route discovery method does not require an updated routing table once constructed. The summary of the MAC protocol and route discovery used in the following studies are listed in Table 2.

5.2. Direct communication of layers

The Network and Physical layer combination automatically uses the direct communication architecture since these layers are not adjacent. This classification varies the modulation technique or exploits the configurable parameters in the Physical layer. At the same time, this classification combines several pieces of information or modifies the Network layer. This combination allows the layers to communicate directly with each other.

The authors of [61] proposed a new modulation technique to solve the issues regarding congestion and scalability in LoRaWAN. This modulation technique, called Dual Orthogonal LoRa modulation, can provide levels of freedom for SF assignments. A cross-layer approach was used to leverage the modulation with a multiplexing technique and ADR adaptation. These processes are done individually by different layers in the stack. However, the authors propose to create a relationship between these layers so that the network performance does not entirely rely on the link but on the network-level performance. The existing ADR and network densification are insufficient to solve the said issues according to the authors of this work. Network densification is the process of adding a gateway near the end devices where areas are suffering from attenuation due to path loss [62]. Applying network densification allows the end devices to utilize a lower SF. Based on real-life water metering application that employs ADR and network densification, it was observed that the end devices usually utilize SF7, SF8, and SF9 [62]. These low utilization of SF, therefore, contribute to the traffic due to LoRa orthogonality, especially for SF7. The proposed modulation scheme has the same time and energy of sending one dual orthogonal symbol with SF8 (14 bits) compared to traditional LoRa modulation when sending two consecutive symbols with SF 7 (14 bits). This result led to the modification of the ADR, where there will be the same amount of SF7 and SF8, which can lower the traffic in the ALOHA access. In addition, the Dual Orthogonal LoRa modulation was also implemented with time-power multiplexing. The gateway needs to respond to several end devices and, at the same time, abide by DC regulations. With this condition, the bottleneck of the capacity of the network is in the downlink [63,64]. Time-power multiplexing can allow the gateway to transmit up to 6 parallel transmissions simultaneously using different SFs. This can allow the Dual Orthogonal LoRa modulation to enhance its bit rate. The effect of bit-rate enhancement on the gateway is that it can send more acknowledgments with the same transmission time and can lessen the transmission time to approximately 56% compared to the traditional LoRaWAN.

Modifying the LoRa modulation related to this study is uncommon. The cross-layer approach indeed showed favorable results in terms of mathematical analysis. However, the assessment of the proposed modulation scheme should be further extended to support its feasibility.

5.3. Completely new abstraction

Combining all the layers can produce more efficient and definite results. Several designs and interactions between the layers can introduce complexity in the network. In addition, when considering the new abstraction approach, this can completely alter the processes in the system. Therefore, special considerations should be observed.

An example of the new abstraction architecture was the study of [18]. The authors created a simulator that co-optimizes the MAC, Physical, and Network parameters to a more realistic scenario through cross-layer. This study split the framework into three modules: the ED, the Air interface, and the GW. The modules have embedded information containing realistic parameters relevant to the energy consumption assessment in LoRaWAN. The end nodes contain the location information, energy profile, and LoRaWAN parameters. The authors specified that the energy consumption in the end nodes is not limited to transmitting, receiving, and sleeping states. Processes like waking up, setting the radio, and executing MAC-related functionality after receiving the downlink can all contribute to power consumption. For the air interface, a realistic simulation of a path loss, Received Signal Strength (RSS) translated to SNR and collision model. The characteristics of the gateway are similar to the iC880 A LoRa Concentrator because this transceiver can receive multiple LoRa packets with different SFs. The packet length was considered in assessing the energy consumption through its Data Extraction Rate (DER), energy per payload byte, and variance of the varying propagation characteristics. One of the observations in this study found that turning on the ADR feature of LoRaWAN is highly recommended for energy optimization. Also, turning OFF confirmed messages or not requiring the gateway's acknowledgment can contribute to energy consumption. In addition, sending large packets can minimize the average energy consumption per payload byte. Therefore, data for delay-tolerant applications can be accumulated to decrease overheads and minimize channel utilization.

For the authors of [65], it adopted the cross-layer framework of [18] to improve energy efficiency. The authors based their MAC protocol approach on exploiting long preambles and CAD. Preamble sampling can identify the state of the channel [66]. Prolonged preambles detected by the end nodes make the radio turn on as long as possible to receive data. The existing LoRa CAD mode then aids in detecting the presence of the preamble [28]. A listen-before-talk mechanism must be performed before a packet transmission. All the events in the network are scheduled at periodic detection intervals. A long preamble must be sent before the transmission occurs. There are at least two required CADs, which should be within the airtime of a prolonged preamble. The transmission is rescheduled to the next CAD cycle when the previous CAD detects a pending transmit message or activity. The routing process only requires a specific ID and hop count. The GW will periodically send route discovery messages. The end nodes are considered to have a direct connection to the GW when they receive the routing discovery message. The end node then stores the GW ID and hop count. The end node that received that message broadcasts the route discovery message and replaces the ID with its own. A connection can only be established with the other nodes when there is a less or equal hop count and a better SNR. The result of considering longer preambles by utilizing CAD minimized the energy spent in the idle mode of end devices. Through experimentation, the protocol provided a PDR of approximately 73.3% to 99%, considering the end devices' position and environmental factors.

The authors of [67] proposed modifying the protocol FREE [68] to improve the synchronization process of end devices. The FREE protocol proposes a bulk collection of buffered data at a specified time. The authors of [67] aim to create a specific time slot for EDs based on their respective SF to address the collision issues in FREE. In this case, the purpose of assigning the SF is to guarantee that the RSSI is larger than the receiver sensitivity. The SF is based on the received power, receiver sensitivity, and propagation loss. The identified time slot for synchronization tends to reduce collision. This is based on the trigger time in which the EDs need to wake up to send synchronization requests to the gateway, the ToA of the transmitted packet, and the guard time between two consecutive time slots. The EDs can send a synchronization request every transmission round. EDs that fail to synchronize must wait

Table 3
Methods used for SF parameter allocation for dynamic end devices.

Paper	Location	Distance reference
Shayo et al. [67]	Trilateration	RSS
Soy et al. [69]	Trilateration	RSS
LoRa+ [70]	Trilateration	RSS
ND-ADR [71]	Multilateration	TDoA

for the next round. The time slot assigned during synchronization depends on the first synced ED. The farthest ED will be prioritized if more than one ED is synced. On the other hand, the time slot for data transmission depends on the time spent synchronizing the end devices. The slot length is based on the maximum ToA of every SF. The EDs with the same SF are grouped, and end devices with different SFs can share the same time slot since they are orthogonal. Comparing the results of this proposal with other existing algorithms like FREE, it produced favorable results in terms of PDR and energy consumption.

The authors of [69] proposed an adaptive SF allocation scheme where the EDs can decide on several actions based on their SF. The ED can assign itself an SF based on the RSSI received from the GW. In addition, the EDs can select a timeslot based on their SF values. This proposal starts with the GW periodically sending beacons to the EDs. The GW selects the highest SF to reach its maximum reachable distance. A preamble and a fixed payload length comprise the beacon frame. The EDs must synchronize to the GW's clock to join the network. Then, EDs go back to sleep mode after the synchronization process. The EDs must wake up whenever the GW sends a beacon to ensure synchronization. The SF assignment of the EDs is based on the RSSI values coming from the beacon signals. After the SF value assignment, the EDs are clustered in different SFs according to their distance to the GW. The ED detects channel variations in a mobile environment and immediately adapts its SF based on the latest beacon frame. This proposal adopted a distributed channel access method to avoid excessive overheads for the scheduling method. The uplink (UL) channel is divided into non-overlapping time slots. Each time slot is based on its index, and it is assumed there will be different ToA for different SFs. The ED selects the timeslot index for its transmission. The packet transmission from ED to GW occurs on one of the orthogonal SFs. Each channel has six parallel subframes that can be used based on their SF. The results of this study showed above 70% PDR when the uplink duration is 120 s. This proposal also showed favorable results compared to the traditional ADR and ADR+ [29] (static and mobile), except for an uplink duration of 30 s, specifically for static conditions.

The authors of LoRa+ [70] proposed a mechanism to decrease the rate of rejected packets and improve the Packet Error Rate (PER). This proposal uses the second receive slot (RX2) of LoRaWAN for synchronization, pre-determined parameter assignment, and carrying information regarding the data ETA for the EDs. This process sends channel decisions like SF, frequency, and DR to the EDs before uplink transmission. The RX2 will transmit a "Configuration Frame" for Class A and B end devices through flooding and beacons, respectively. The GW assigns an SF to the EDs using the RX2 slot after the channel estimation and synchronization are completed. The GW informs the ED to adopt a new SF based on the estimated RSSI if the same SF is used during consecutive transmissions. The opening of the RX1 slot depends on the ACK in the network. The use of acknowledgment requires the ED to listen during RX1. LoRa+ confirmed the decrease of packet collision and rejected packets through simulation. LoRa+ has a lesser PER compared to traditional LoRaWAN. Moreover, the protocol can also reduce the use of gateways up to 80% with high-density end devices, making the system more cost-efficient.

The authors of [71] propose a mechanism called ND-ADR. This process addresses data collision when the EDs use the same parameters during transmission. According to the authors, the slow adaptation of EDs to the parameter allocation of the ADR causes destructive concurrent transmission in the network. Destructive concurrent transmissions occur when two simultaneous or slightly overlapping transmissions use the same parameters. A nondestructive concurrent transmission time is assigned to the EDs. The transmission time is based on the previous transmission parameters of the EDs. The ND-ADR works in three phases: maintaining the arrival time of consecutive transmissions, end device profiling, and the assignment of nondestructive transmission time. The objective of the first phase is to pair the collected arrival time of packets sent by the EDs for every regular UL. In the second phase, the NS places the EDs in groups based on their ID, SF, and transmission time. This aims to find concurrent transmissions using the same SF in the same channel. Lastly, the nondestructive transmission time assignment will be based on the minimum and maximum margin between the two concurrent transmissions. The data can be demodulated successfully if the margin between two simultaneous transmissions exceeds their ToA. The NS calculates the suitable SF, TP, and nondestructive transmission time and assigns them to the EDs. Compared to the traditional ADR and modified ADRs [72,73], the ND-ARD performs better in terms of energy consumption and PSR (when both ED and NS receive DL (ACK) and UL packets, respectively, in one of the available receive windows).

The work of [18] introduced flexibility in the network design by adopting the completely new abstraction cross-layer approach. Most of the frameworks and simulators created for LoRa do not support the different cross-layer approaches. Therefore, the proposed framework can be further exploited and used as a reference for future cross-layer designs. For example, the cross-layer proposal of [65] is based on this framework and showed favorable results.

The goal of dynamic SF parameter allocation to EDs is the focus of the authors of [67,69,70], and [71]. The SF allocation can be done by predicting the distance between the EDs and the GW. The distance can be predicted by using trilateration or multilateration. Trilateration uses the Received Signal Strength (RSS), while multilateration uses the Time Difference of Arrival (TDoA) to predict the distance between the ED and the GW. The summary of methods used for identifying the distance of the ED to GW of the following proposals is shown in Table 3.

5.4. Shared database

Creating a relationship between the link-level metrics, traffic user parameters, and network operation is the goal of [74]. The core idea of this proposal is to relate several parameters in different layers where they can be controlled in a specific database. This tends to create a model that can account for the optimized parameters in the network and the distribution function. The model is constructed based on network parameters like SF, data rate, BW, and the different probabilities of data delivery. The proposed system comprises a network, a channel, and a traffic model. The network model can control several tasks and interactions of the GWs and EDs. The controller used in the network model is based on a Software Defined Network (SDN) architecture. The SDN process logically controls the EDs by separating the control and data plane [75]. The purpose of the control plane is to manage the data, while the data plane is for moving the data. The channel model is based on the initial received network parameters, the organization of the data collection cycle, and the estimation of the parameter values to be transferred to the EDs. The traffic model can assume two options: deterministic flow of data and event-driven data. The implementation of this protocol showed favorable results in terms of collision, frame loss, and delivery probabilities when compared to the work of [76,77], and [78].

One of the features of eLoRa [79] is jointly optimizing the parameters in the Physical and Link layers. This proposal uses a cross-layer approach to assign the optimized parameters from the PHY and LINK layers to the EDs. Assigning a higher SF is more reliable in terms of transmission. However, the data rate decreases, leading to additional energy consumption. The authors of eLoRa first identified the effects of SF and packet length on PER. For low SNR scenarios, a change of packet length from 80 bytes to 40 bytes can decrease the PER from 0.4 to 0.3, respectively. The PER is also reduced from 0.4 to 0.2 for SF7 to SF12. Directly jointly tuning these parameters may lead to a better QoS. However, it can drastically affect energy utilization. Therefore, a cross-layer framework was proposed considering the high-level requirements of lifetime, range, and latency. The eLoRa framework has a process where gateways are connected to a cloud service. In this cloud, one of the processes is parameter optimization. The cross-layer parameter optimization was done through high-level programming. The authors solved the optimization problem using an integrated development environment software called Ecross-layeriPSe. The optimized parameters depend on the network model and monitor the state of the network. The number of data packets to be merged, block size, and the configurable parameters of the PHY layer are included in the optimization process. These parameters are then related to high-level metrics like lifetime, reliability, and latency. The gateway relays these parameters to the EDs. The parameters of the end devices are then updated depending on the commands sent by the gateway. This protocol achieved a 55.7% additional communication range, a packet reception ratio 60% higher, and a node lifetime 46.6% higher compared to traditional LoRaWAN.

The proposals of [74,79] provided more detailed and comprehensive results in optimizing the LoRaWAN network. However, these processes introduced complexity to the design compared to the other cross-layer architectures.

Based on the reviewed papers, cross-layer optimization can enhance the performance of LoRaWAN. Table 4 summarizes the classification of the cross-layer approaches in LoRaWAN. The criteria used in the classification are the protocol application, classification of the cross-layer architecture, assessment, the layers utilized in the proposals, and the purpose of crossing the layers. The detailed performance of cross-layer optimization for LoRaWAN will be further discussed in the next section.

6. Performance of cross-layer-based optimization approaches in LoRaWAN for IoT

This section presents the performance and effectiveness of the different cross-layer approaches in optimizing LoRaWAN. This discusses the impact of the cross-layer approach in solving energy consumption, communication range, and multiple access challenges.

6.1. Energy consumption

Using different cross-layer approaches can address energy consumption in many ways. The solutions can be through a scheduling and routing technique, parameter allocation, or different cross-layer optimization approaches.

- *Scheduling and Routing*: The authors of [28,48,51,56], and [60] merged the MAC and NET layers of LoRaWAN to address energy consumption through proper node scheduling and route discovery. If assessed individually, the reservation-based MAC protocol in the Link layer requires information regarding the network to establish a schedule. In addition, synchronization between the GW and EDs must be accurate. These processes incur large overheads [36]. This is the same with the routing in the Network layer, where most proposals have considered proactive route discovery. This type of routing protocol requires the routing tables to be updated often. With the use of cross-layer, these processes are merged to work together to limit the overheads in the network.
- *Parameter allocation*: The SF is one of the most important LoRa parameters. Therefore, several cross-layer proposals tried to assign an efficient SF to the EDs depending on their traffic condition or location. The authors of [61] proposed to balance the traffic in terms of SF, especially with low SFs, based on a modified ADR algorithm. The authors of [67,69,70], and [71] proposed different methods to allocate an efficient SF parameter to the EDs dynamically. Methods like trilateration and multilateration can be used to identify the distance between the ED and GW. The trilateration method uses the ToA or RSS to determine the distance between the transmitter and the receiver. Usually, this method uses a path loss model to predict the RSS value. The multilateration method usually uses two synchronized transmitters and computes the TDoA of data when it reaches the ED [80]. Imagine an end node receiving a LoRa signal to at least two GWs. Using trilateration requires synchronization

Table 4
Classification of cross-layer optimization approaches in LoRaWAN for IoT.

Paper	Assessment				
	Application	*Architecture*	*Validation*	*Layers involved*	*Goal*
LoRaBlink [28]	Generic IoT	Merging of layers	Testbed	Network and Link	Multi-hopping with node scheduling
JMAC [48]	Smart cities	Merging of layers	Testbed	Network and Link	Multi-hopping with node scheduling
Mai et al. [51]	Suitable for agriculture and tunnels	Merging of layers	Testbed	Network and Link	Multi-hopping, node scheduling, and controlling packet size
Dias et al. [54]	Generic IoT	Merging of layers	Testbed	Network and Link	Route construction based on the end nodes interaction
OppLoRa [56]	Generic IoT	Merging of layers	Simulation	Network and Link	Improve routing through the broadcast nature of the multiple transmission/return packet from MAC to network to select different node for re-transmission
IRIS [60]	Pipeline type network topology	Merging of layers	Simulation and Testbed	Network and Link	MAC layer to restrict node transmission for network efficiency
Vangelista et al. [61]	Generic IoT	Direct communication	Mathematical	Network and Physical	SF freedom assignment and Time Power Multiplexing
Callebaut et al. [18]	Generic IoT	Completely new abstraction	Simulation	Network, Link, and Physical	Co-optimize layers to realistic conditions
Leenders et al. [65]	Wide-spread and ultra-low throughput WSN	Completely new abstraction	Testbed	Network, Link, and Physical	Combine prolonged preamble and CAD implemented in the network layer
Shayo et al. [67]	Generic IoT	Completely new abstraction	Simulation	Network, Link and Physical	Create time slots based on the SF
Soy et al. [69]	Generic IoT	Completely new abstraction	Simulation	Network, Link, and Physical	Dynamic SF allocation and clustering of EDs
LoRa+ [70]	Generic IoT	Completely new abstraction	Simulation	Network, Link, and Physical	Control parameter allocation and scheduling
ND-ADR [71]	Smart city	Completely new abstraction	Simulation	Network, Link, and Physical	Scheduling of new transmission time depends on TDoA and dynamic SF allocation
Muthanna et al. [74]	Generic IoT	Shared database	Mathematical	Network, Link, and Physical	Analytical relationship between quality of the network, link level, and traffic parameters
eLoRa [79]	Generic IoT	Shared database	Testbed	Network, Link, and Physical	Optimize SF and packet length simultaneously

between the transmitter and the receiver. For multilateration, the transmitters are synchronized, whereas the receiver does not need to be. Based on several studies, the use of TDoA is more accurate than the use of RSS [80]. A possible reason is that RSS-based reading needs to consider an accurate path loss propagation model. In addition, cross-layer was also used in some of the proposals to cluster or group the EDs depending on their assigned SFs to optimize the energy consumption during transmission.

- *Effects of using different cross-layer architectures for energy consumption*: Most of the surveyed papers proposed different cross-layer approaches to optimize the energy consumption of LoRa EDs. The freedom of selecting the parameters and layers in the protocol stack creates diverse solutions regarding cross-layer optimization. The proposals of [18,65] considered all the processes related to energy consumption in the entire LoRaWAN network. Some proposals used different cross-layer approaches and methods but had similar results. For example, the work of [28,65] employed different cross-layer approaches with similar performance metric evaluation. The study of [28] merged the Network and Link layer to efficiently manage the energy of end devices through multi-hop and node scheduling. On the other hand, the authors of [65] utilized a new abstraction using all the layers to address the same issue using long preambles and CAD. Results showed that the work of [28] had an 80% PDR, while the work of [65] had an average of 86% PDR. These two studies are insufficient to come to a particular conclusion. However, it can be considered that different cross-layer architectures and approaches can lead to similar results. This also shows the diversity of possible solutions in using different cross-layer approaches. In addition, based on the summary of results, the use of cross-layer is evident in optimizing the energy consumption of the LoRaWAN network. The summary of the performance of the cross-layer approaches related to energy consumption is shown in Table 5.

Table 5
Performance of cross-layer optimization approaches related to energy consumption.

Paper	Metrics	Parameters	Summary of results
LoRaBlink [28]	Packet Delivery Ratio (PDR)	Energy consumption, Communication range	Packet delivery of nodes up to 80% reliability
JMAC [48]	PDR, End-to-End delay (E2E), throughput	Energy consumption, Communication range	Energy consumption in nodes increases approximately twice when the payload is doubled/energy consumption in nodes is higher when more nodes are connected as child/siblings energy consumption in leaf nodes are lower since it does not transmit/receive often
Mai et al. [51]	Window size, PDR, E2E delay	Energy consumption, Communication range	Packet delivery of nodes up to 97.6%/decreased time slots and E2E delay due to parallel transmissions
OppLoRa [56]	E2E packet error ratio, E2E delay, channel usage (accumulated airtime divided by the duration of simulation)	Energy consumption	Outperforms traditional LoRa by one magnitude in terms of E2E PER
IRIS [60]	Throughput and delay (monitoring information to reach the base station after route establishment), robustness to unreliable wireless connections	Energy consumption	0.35% transmission of route node duty cycle and 0.47% with extra ACK for report
Vangelista et al. [61]	Bit Error Rate (BER)	Energy consumption, Multiple access	70%–80% link-level bit rate increase (incurring 3 dB SNR penalty)
Callebaut et al. [18]	Data Extraction Rate (DER), energy per payload byte	Energy consumption	Approximately 80% decrease in energy per byte when ADR is ON and NO CONF
LoRaBlink [28]	Packet Delivery Ratio (PDR)	Energy consumption, Communication range	Packet delivery of nodes up to 80% reliability
Leenders et al. [65]	PDR	Energy consumption	An efficient PDR of approximately 73.3% to 99% depending on the position of the end devices and environmental factors
Shayo et al. [67]	PDR and energy consumption (J)	Energy consumption, Multiple access	3.65 J average energy consumed depending on the number and range of EDs/average of 45% PDR greater than FREE depending on the number and range of EDs/average
Soy et al. [69]	PDR	Energy consumption	Above 70% PDF for certain scenarios/outperformed traditional ADR and ADR+ for certain scenarios
LoRa+ [70]	Packet Error Rate (PER), rejected packet rate (based on channel sensitivity)	Energy consumption, Multiple access	Approximately 75% and 70% less rejected packets compared to traditional LoRaWAN when 1 GW is used in small cities and rural areas respectively and much better results when using 2 GW's compared to LoRaWAN that used 6 GW's. 60% PER for 150 EDs compared to traditional LoRaWAN that produced the same PER in 100 EDs
ND-ADR [71]	Packet Success Ratio (PSR), energy consumption (J)	Energy consumption	ND-ARD (200-1000 EDs): approx of 0.75 J to 2.5 J for RX1, 0.55 J to 1.6 J for RX2; Traditional ADR (200-1000 EDs): approx of 1 to 10.5 J RX1, 0.9–10 J RX2
eLoRa [79]	Lifetime (duty cycle, battery capacity, current drawn from transmitting, receiving, idle), Range (distance of ED to GW, PRR)	Energy consumption, Communication range	60% PRR higher and 46.6% longer lifetime compared to traditional LoRaWAN

6.2. Communication range

Several papers proposed co-optimizing energy consumption and communication range in the LoRaWAN network. The traditional LoRaWAN network usually employs a star topology. The EDs are directly connected to a concentrator or GW (one-hop) in this topology. The disadvantage of this topology is that the entire network will be lost when the concentrator fails. In addition, EDs that are farther from the GW consume a higher TP. Most solutions to extend the communication range and simultaneously address energy consumption are through the multi-hopping of end nodes. The end nodes can send or route data until it reaches the destination. Depending on the routing protocol used, this allows the end nodes to have a shorter transmission path, thus consuming less energy compared to a single hop. The routing in a non-cross-layer approach essentially has the same operation compared to the papers presented. The only difference is that the scheduling is integrated into the routing process.

6.3. Multiple access

Based on the reviewed papers, the identified cross-layer solutions to multiple access challenges involve reservation-based MAC protocols. The MAC layer usually uses a TDMA-based scheme, where specific time slots are assigned to the EDs to access the medium to avoid collision. These time slots can also provide fairness in all the EDs and cater to additional EDs in the network, depending on the schedule created. Properly addressing factors related to network transmission can also optimize multiple access challenges. The authors of [74] proposed a method to optimize the access network parameters by knowing the probability of collision, frame loss, and data delivery. This solution is based on a model that can collaboratively manage the layers in the stack.

The following cross-layer optimization approaches would not be possible if the OSI model is adapted. There are no known standards for using cross-layer optimization. This optimization process allows users to treat the layers however they desire. The reviewed papers showed that cross-layer optimization can solve issues regarding LoRaWAN challenges in IoT applications. The next section discusses the following issues and challenges of using a cross-layer approach optimization technique.

7. Issues and challenges

This section discusses the following challenges and potential solutions when implementing cross-layer optimization in LoRaWAN for IoT. There are several issues when utilizing cross-layer optimization in wireless networks. Using this optimization technique should consider the interoperability, compatibility, sustainability, and support of a non-complex structure as much as possible. A summary of issues and potential solutions can be seen in Table 6.

1. *Standardization:* There are no restrictions or prescribed processes for utilizing the cross-layer approach. The freedom to create any combination or use any architecture, however, can lead to non-uniformity and a lack of organization that can hinder the future use and evolution of the system. In addition, the lack of standardization in a process can increase the probability of losing control and produce uncertainties in the system. The cross-layer optimization approach is not a new process but is still used as of this writing for wireless optimizations. Therefore, an updated survey, studies, and tutorials regarding this matter can contribute to solving this issue.
2. *Compatibility and Trade-offs:* Cross-layer combinations are non-adjacent to each other. These layers have no co-relation in the traditional protocol stack. Crossing these layers should assess compatibility, sustainability, and fairness to the network. Therefore, having a protocol that can guarantee that the operations in these respective layers can be harmonically combined is essential. Incorporating computing and control mechanisms is a possible solution. For example, in [74,79] an SDN controller and a cloud service for high-level computations were added, respectively.
3. *Lacking assessment approaches and troubleshooting:* The optimization processes lack real-life implementation. In addition, the testbed implementations were evaluated using small-density networks. As a result, there may be discrepancies when deployed in actual and diverse scenarios. Proper documentation and complete mathematical models can be a solution to validate the proposals thoroughly. On the other hand, a protocol that is scalable can also be a solution for experiments done with small-density networks. A possible solution is to create a cross-layer simulation tool that can be tuned and adapt cross-layer functions inspired by the framework called NS-MIRAcross-layerE [81] and the study of [18].
4. *LoRaWAN Class C:* The majority of the devices used in optimizing LoRaWAN were Class A and B devices. LoRaWAN optimizations seldom use Class C devices due to their high energy consumption. These end devices do not often rely on batteries and are mainly powered. This specification can be used for time-critical IoT applications or systems that require the end devices to stay awake the entire time. This can also exploit the scheduling process of a specific protocol and introduce simplicity in the design. Lastly, explore the operation of Class C end devices in the 2.4 GHz ISM Band, which no longer requires a duty cycle restriction [82].
5. *Dynamic End Devices:* The papers in this study assessed the end devices in a static topology. There are now studies trying to address mobility problems in LoRaWAN. These studies used different approaches, like re-configuring the Adaptive Data Rate (ADR) [69,83–85], studies suggesting a mobile gateway [86–89], and machine learning [90–95]. However, these studies are not focused on cross-layer optimization. There are cross-layer studies indicating that this technique can address issues about mobility. The authors of [96,97] proposed that network and link layer interactions can solve mobility problems. In addition, the coordination of lower layers to higher layers can provide better transmission of data and reduce the effect of mobility [98]. A potential solution is to consider all the layers in the stack and incorporate them with other processes like SDN, fog computing, machine learning, and reinforcement learning. Another potential solution is exploiting OR, where routing can adapt to the dynamic situation of the end devices [58].

Addressing these challenges can minimize uncertainties in the system since the cross-layer approach is not a standard protocol. In addition, complete documentation and new insights can help in the evolution of cross-layer optimization in LoRaWAN for IoT. The standardization of processes, compatibility, and trade-offs, and the assessment of dynamic end devices are the most important issues that should be considered. Since cross-layering processes are not similar to the accustomed traditional stack, there may be issues regarding the system's sustainability. In addition, the effectiveness of the presented solutions may not apply to dynamic end devices.

Table 6
Potential solutions to issues and challenges.

Issue	Possible solution
Standardization	Up-to-date studies, surveys, and documentation regarding cross-layer approaches
Compatibility and Trade-offs	Incorporation of computing and controlling mechanisms
Lacking assessment approaches and troubleshooting	Process of proposal validation and simulation tools that can support the cross-layer architecture and LoRaWAN specifications
LoRaWAN Class C	To be used in time-critical and non-delay tolerant IoT applications/exploration in the 2.4 GHz ISM band
Dynamic end devices	Consideration of all the layers and incorporating other mechanisms to produce adaptive protocols

8. Future research directions

The IoT has become a ubiquitous system and still creating more opportunities and innovation in the field of technology. However, this development simultaneously increases the challenges in the said area. To keep up with this growth and changes, the potential future directions of Cross-Layer Optimization for LoRaWAN in IoT are the following:

1. *Adaptive protocols for mobility:* The IoT is diverse and being considered in many applications. Several uses of IoT are towards Unmanned Aircraft Systems (UAS), Search and Rescue (SAR), Wireless Body Area Networks (WBAN), satellite communications, and end devices that are moving. As discussed in the previous sections, there are still issues regarding the mobility of end devices. Some studies are directed toward the mobility of end devices in LoRaWAN. However, these studies have not yet further explored cross-layer optimization. Therefore, one possible future direction is to consider this approach and gauge if it is efficient for dynamic environments. Several routing protocols like Opportunistic Routing (OR) and processes used in VANETS like routing, scheduling, and localization should be considered.
2. *Heterogeneity:* The exploitation of LoRaWAN made it feasible to be used in other applications aside from long-range communications. Several studies proposed interconnecting different platforms with LoRaWAN as its catalyst because of its unique qualities. For example, the study of [99] used LoRaWAN as a bridge for different wireless platforms to interconnect other wireless technology. In line with this, the freedom of cross-layer optimization and openness of LoRaWAN can be further explored for cross-technology collaboration with other IoT technologies.
3. *Collaboration with other processes:* Studies incorporating cross-layer optimization with intelligent systems like in the study of [100–102], and [103] are becoming popular in wireless communication systems. Several studies incorporated cross-layering and LoRaWAN with other processes like SDN and cognitive radio [74]. In addition, reinforcement learning is now being used in LoRa systems for optimization similar to [104,105]. In addition, reinforcement is now being exploited in cross-layer optimization like in [106] in IoT applications. In line with this, addressing LoRaWAN challenges by collaborating cross-layer approaches with intelligent computing systems can be considered.
4. *Consideration of other parameters:* Most optimization processes in LoRaWAN are either modifying the scheduling, routing, or modulation process. Studies explore communications with confirmed messages [107] and overheads [108] for improving LoRaWAN. With these proposals, future cross-layer optimization in LoRaWAN research can explore other properties or variables to prolong the network's lifetime and improve QoS. In addition, to preserve the original qualities of LoRaWAN to keep the system simple.
5. *Self-sustaining system:* Several IoT developments are considering green technology in their designs. Possible future research can explore a sustainable nature-inspired system to make LoRaWAN more efficient and feasible for optimum QoS without restricting its data bandwidth and transmission power, while at the same time not compromising other essential parameters.

The possibility of integrating intelligent mechanisms with cross-layer optimization and the consideration of green technology or nature-inspired approaches to LoRaWAN can lead to innovation and sustainable communication. Collaborating with high-level and powerful mechanisms with a cross-layer approach can increase the precision and decision-making of the system.

9. Conclusion

This paper presents a review of the different cross-layer-based optimization approaches in LoRaWAN for IoT networks and applications. The 11 research articles reviewed in this study came from IEEE, ScienceDirect, MDPI, and ACM Digital Library of which the majority are Scopus-indexed and Web of Science-indexed. The papers were analyzed according to their prescribed application, cross-layer architecture, the combination of layers, protocol evaluation, and the target optimized parameter. The performance of each approach was analyzed in terms of energy efficiency, communication range, and multiple access. Moreover, the impact of cross-layer-based optimization in LoRaWAN and the observation of the topology of end devices in the studies were added. Lastly, issues, challenges, possible solutions, and future research directions for cross-layer-based optimization in LoRaWAN for IoT were presented in this work.

The collision of data, packet length, bit rate, node scheduling, transmission parameters, duty cycle, and channel sensitivity can contribute to the energy consumption of LoRaWAN. Therefore, the majority of the optimization process efficiently enhanced the energy consumption of LoRaWAN. The cross-layering approach can optimize energy consumption individually or collaboratively with other parameters like communication range and multiple access. It was also observed that the protocols were evaluated with static end devices. Hence, the protocols may not be as effective or feasible when applied to dynamic end devices.

The utilization of cross-layer-based optimization for LoRaWAN should consider the interoperability, compatibility, troubleshooting, and simplicity of the system. Taking into account the following considerations can minimize the possibility of uncertainties during run time. The cross-layer-based optimization can also collaborate with other computing and intelligent systems. There are studies included in this paper that combined cross-layering with Software Defined Networking (SDN) and Cognitive Radio (CR). In line with this, a possible future direction is combining cross-layering with intelligent systems. For example, using machine learning to predict parameters for protocols that can adapt to the dynamic changes in the environment.

CRediT authorship contribution statement

Melchizedek Alipio: Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization, Investigation, Validation, Writing – original draft. **Carl Christian Chaguile:** Writing – original draft, Visualization, Formal analysis, Data curation. **Miroslav Bures:** Writing – review & editing, Resources, Funding acquisition, Formal analysis.

Declaration of competing interest

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Data availability

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